Alexander Arens

Dallas, TX 75225 | 214-289-0958 | alexander.arens04@gmail.com

EDUCATION

Purdue University — **West Lafayette, IN**
Bachelor of Science, Electrical and Computer Engineering — **December 2027**

PROFESSIONAL EXPERIENCE

Fuse Medical — **Dallas, TX**
Engineering Intern — **May 2024 – Present**

- Reverse-engineered the K-wire alignment system from a newly acquired company to create a CAD file.
- Collaborated with the lead engineer to develop initial prototypes of tibial baseplates and inserts. Designed and prototyped osteotomy and general fusion plates for a foot plating system.
- Redesigned SolidWorks CAD templates to improve efficiency and standardization.
- Maintain remote work arrangements and supply prototype components.

PERSONAL PROJECTS

- Full portfolio link https://flukeychip.github.io/Project-Portfolio/
- **Scholarship Recipient – All-In Summit:** Selected as a scholarship recipient for the All-In Summit in California from a pool of thousands of applicants. Engaged with leaders in technology, medical, business, and government sectors. Invited to contribute to a medical AI startup.
- **6DOF Arm:** Drawing inspiration from the robotic arms available on campus, I have spent several weeks designing my own. The project is still in the CAD and design phase, but it is likely to my biggest project to date.
- **Large format laser engraver:** Designed and built a large-format (500 mm x 500 mm) laser engraver featuring automatic homing functionality and a two-factor safety system.
- **Rapidly deployable quadcopter:** Led a team to develop a drone with a compact, easily manufacturable design, a hot-swappable camera system, and ultrasonic sensor-based obstacle avoidance.
- **Voron .01:** Developed a 3D printer featuring a belt-driven Z axis and integrated vibration compensation to enhance performance and reliability. Upgraded firmware to Klipper and engineered a lightweight, simplified hot end.
- **Smart Drink Dispenser Table:** Developed a device designed for discreet integration into a tabletop. Engineered the system to autonomously prepare beverages in response to user commands.
- **High current USB C power adapter:** Designed a circuit to request 20v 3A from a USB C PD device an convert it to 5V 12A to power components that require high current.
- **Automated home security:** Lead a team of two other engineers to design a pan tilt camera system with IR flashlights, lasers and cameras to track and detect humans within a 360 radius. Extensively used cursor to program image recognition with closed loop visual feedback including PD control adaptive dead zones and platform specific optimizations
- **Monodirectional speaker:** Designed a circuit to drive ultrasonic speakers at 40 kHz and modulate them with an audible carrier wave. Produces directional audio output, audible only when the speaker is pointed directly at the listener.
- **Active orthosis (ASME):** Collaborated with the electronics division of an active orthopedic device that allowed patients with nerve or muscular damage to arms to restore movements using a multitude of sensors and actuators

RELEVANT SKILLS

- **Mechanical & CAD:** Proficient in Fusion360, SolidWorks, NX, KiCAD, and Autodesk Eagle.
- **Electronics:** PCB design, microcontrollers (ESP32, ATmega, RP2040), analog/digital analysis
- **Software & Programing:** FPGA (Verilog and ICE), MATLAB, Microsoft Excel, C, C++, Python
- **Prototyping:** Additive manufacturing (FDM, SLA, SLS), CNC machining, laser cutting, MIG welding

LEADERSHIP AND INVOLVEMENT

Eagle Scout | Phi Delta Theta project manager, Assistant House Manager | 3D Printing Club | PSP active controls | Purdue ASME